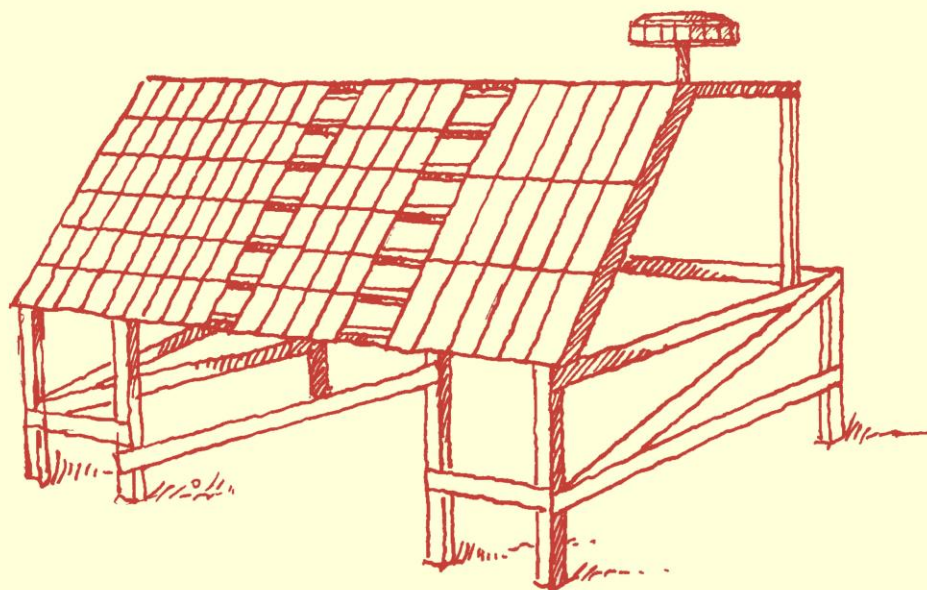


CONVENTION ON LONG-RANGE TRANSBOUNDARY AIR POLLUTION

UN/ECE INTERNATIONAL CO-OPERATIVE PROGRAMME
ON EFFECTS ON MATERIALS, INCLUDING HISTORIC
AND CULTURAL MONUMENTS



Report No 72:

Results of corrosion and soiling from the 2011–2012
exposure programme for trend analysis.

September 2013

PREPARED BY THE MAIN RESEARCH CENTRE

swerea | **KIMAB**

Swerea KIMAB AB, Stockholm, Sweden

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Results on corrosion and soiling from the 2011–2012
exposure programme for trend analysis.

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1 Introduction

Early in the discussions on the “Convention on Long-range Transboundary Air Pollution” it was recognized that a good understanding of the harmful effects of air pollution was a prerequisite for reaching agreement on effective pollution control. To develop the necessary international cooperation in the research on and the monitoring of pollutant effects, the Working Group on Effects (WGE) was established under the Convention in 1980 and held its first meeting in 1981. The Convention involves countries in the UNECE region and has its secretariat with the UNECE [1].

The Working Group on Effects provides information on the degree and geographic extent of the impacts on human health and the environment of major air pollutants, such as sulphur and nitrogen oxides, ozone and heavy metals. Its six International Cooperative Programmes (ICPs) and the Task Force on Health identify the most endangered areas, ecosystems and other receptors by considering damage to human health, terrestrial and aquatic ecosystems and materials. An important part of this work is long-term monitoring. The work is underpinned by scientific research on dose-response, critical loads and levels and damage evaluation [1].

The International Co-operative Programme on Effects on Materials including Historic and Cultural Monuments (ICP Materials) is one of the six ICPs. During the course of the programme, which had its first exposure in 1987, the special importance of trend effects has become more and more evident. The aims of ICP Materials are now to

- perform a quantitative evaluation of the effects of multi-pollutants such as S and N compounds, ozone and particles as well as climate parameters on the atmospheric corrosion of important materials, including materials used in objects of cultural heritage;
- describe and evaluate long-term corrosion trends attributable to atmospheric pollution in order to elucidate the environmental effects of pollutant reductions achieved under the Convention and in order to identify extraordinary environmental changes that result in unpredicted materials damage;
- use the results for mapping areas with increased risk of corrosion, and for calculation of cost of damage caused by deterioration of materials.

The present report addresses the second of these aims and gives the results of corrosion from the most recent (2011-2012) exposure programme for trend analysis. Complete reporting of environmental data from this exposure and, as a consequence, an analysis of trends in corrosion, soiling and pollution and their relation, will be reported in 2014. Here quantified trends in key selected parameters based on the Guidelines on Reporting of Monitoring and Modelling of Air Pollution Effects are reported (carbon steel, zinc, limestone), as well as quantified trend of copper, weathering steel and soiling of modern glass.

2 Corrosion

2.1 Carbon steel

Carbon steel was prepared and evaluated by the sub-centre in Czech Republic (SVUOM).

Plates of unalloyed carbon steel (with C < 0.2 %, P < 0.07 %, Cr < 0.07 % according to CSN 11373) with dimensions 100 x 150 x 0,5 mm³ were used, triplicate per exposure period.

Before the exposure the samples were degreased in alkaline degreaser, rinsed with ethanol, dried and weighed.

The basic information about corrosion behaviour of carbon steel in different environmental conditions gives the value of mass change and corrosion loss. After exposure samples were weighed (conditioned in laboratory conditions) and the corrosion products were removed using the pickling method according to ISO 8407 (Annex A - solution C3.5). In consecutive pickling cycles the corrosion layer was completely removed without dissolving the base metal. The repetition of the procedure stopped if the surface appears clean. The corrosion losses had been obtained by gravimetric method.

Corrosion losses for individual samples of carbon steel exposed in open atmosphere for 1 year exposure (2011/12) are summarised in Table 1 and presented in Figure 1. The test sites with the greatest corrosion losses of samples were No 3 (Kopisty), No 10 (Bottrop) and No 50 (Katowice) where industrial pollution still exists. The corrosivity is in category C2 for carbon steel but with the three exceptions above in C3. In Figure 2 the trend in yearly corrosion loss of carbon steel since 1987 is presented. At test sites located in industrial areas the yearly corrosion mass loss of carbon steel increased somewhat in the last years' corresponding to a small increase of SO₂ pollution level.

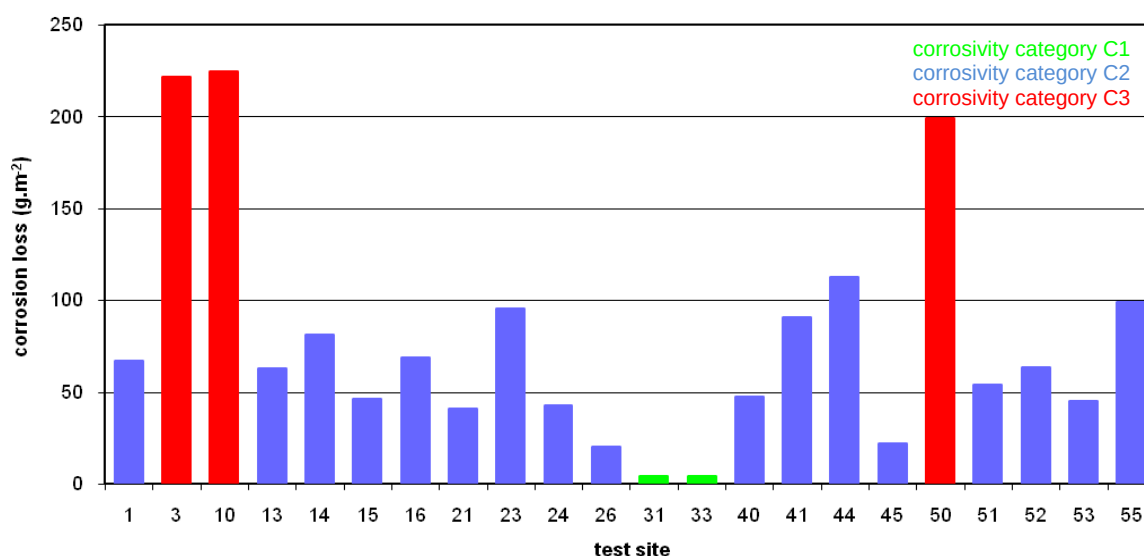


Figure 1 – Corrosion loss of carbon steel after 1 year exposure (2011-2012)

Table 1 - Corrosion losses (g.m⁻²) of carbon steel after 1 year exposure in open atmosphere (2011-2012)

Test site	ML ₉₁ (g.m ⁻²)	ML ₉₂ (g.m ⁻²)	ML ₉₃ (g.m ⁻²)	ML (g.m ⁻²)	CL (μm)
1 Prague	68,9100	67,3767	64,3433	66,8767	8,51 C2
3 Kopisty	225,8867	217,2733	222,5300	221,8967	28,23 C3
10 Bottrop	225,9567	225,4400	221,9967	224,4644	28,56 C3
13 Rome	59,1133	62,2667	66,6300	62,6700	7,97 C2
14 Cassacia	77,4367	78,8333	86,7067	80,9922	10,30 C2
15 Milan	44,0567	44,1067	49,6933	45,9522	5,85 C2
16 Venice	68,8167	68,6533	69,5200	68,9967	8,78 C2
21 Oslo	37,4200	40,3967	45,4833	41,1000	5,23 C2
23 Birkenes	94,6133	98,8833	92,0033	95,1667	12,11 C2
24 Stockholm South	44,3300	41,4767	42,1533	42,6533	5,43 C2
26 Aspvreten	18,1633	20,8900	21,0367	20,0300	2,55 C2
31 Madrid	4,2633	4,2267	4,3900	4,2933	0,55 C1
33 Toledo	4,0300	4,1733	4,1500	4,1178	0,52 C1
40 Paris	41,7233	49,4833	51,5267	47,5778	6,05 C2
41 Berlin	94,3333	88,7600	88,3533	90,4822	11,51 C2
44 Svanvik	116,6000	109,4033	111,8467	112,6167	14,33 C2
45 Chaumont	22,3867	20,7833	22,0267	21,7322	2,76 C2
50 Katowice	200,8067	198,4467	198,6333	199,2956	25,36 C3
51 Athens	56,3100	51,8067	53,0167	53,7111	6,83 C2
52 Riga	59,9400	64,8667	65,4167	63,4078	8,07 C2
53 Vienna	44,7433	46,8867	43,4900	45,0400	5,73 C2
55 Sankt Peterburg	114,0000	85,2733	97,4167	98,8967	12,58 C2

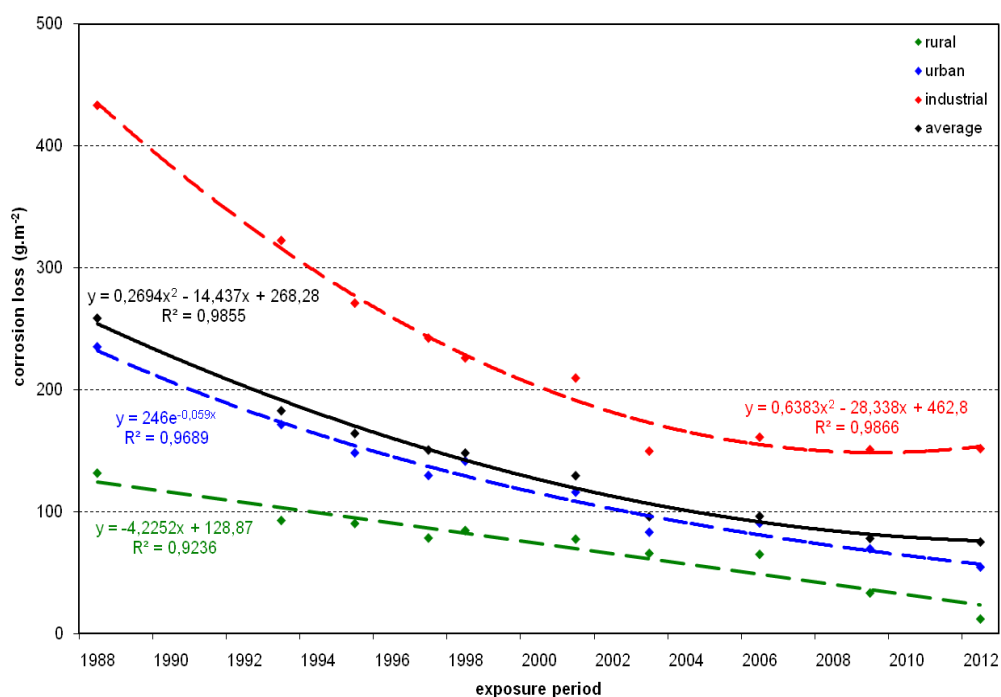


Figure 2 – Trend in yearly corrosion loss of carbon steel (selected test sites)

2.2 Zinc

The 2011-2012 exposure of zinc was co-ordinated by the sub-centre in Switzerland (EMPA). Plates of zinc (99.99%) with dimensions of 100 x 150 x 2 mm³ were used, triplicate per exposure period. Before exposure the samples were degreased with petrol ether, glass blasted resulting in a surface roughness of (R_a) about 2.9 μm . Pickling after exposure was done with concentrated glycine solution.

Corrosion losses for individual samples of zinc exposed in open atmosphere for 1 year exposure (2011/12) are summarised in Table 2. The highest values are observed in Kopisty, Bottrop, Birkenes, Aspvreten and Katowice. Birkenes and Aspvreten, rural sites in Norway and Sweden, are exceptions when comparing to the carbon steel results and point to a confounding factor specific for zinc. Present data points in the direction of organic acids but this will be discussed further in the 2014 report.

Table 2 Corrosion losses (g/m^2) of zinc (CH) after 1 year exposure in open atmosphere; Trend exposure 2011/12

Test site	ML ₁ (g/m^2)	ML ₂ (g/m^2)	ML ₃ (g/m^2)	ML ^{a)} (g/m^2)	SD ^{b)} (g/m^2)
1 Prague	5.84	5.94	5.94	5.91	0.05
3 Kopisty	9.31	9.90	10.89	10.03	0.65
10 Bottrop	7.60	7.57	7.48	7.55	0.05
13 Rome	3.80	3.75	3.74	3.76	0.03
14 Cassacia	2.34	2.34	2.26	2.31	0.04
15 Milan	6.00	5.98	6.17	6.05	0.09
16 Venice	5.05	4.47	4.84	4.79	0.24
21 Oslo	7.04	7.11	7.28	7.14	0.10
23 Birkenes	6.63	6.83	6.70	6.72	0.08
24 Stockholm South	3.72	4.13	3.96	3.94	0.17
26 Aspvreten	7.44	7.23	6.64	7.10	0.34
31 Madrid	4.48	3.69	4.00	4.06	0.33
33 Toledo	5.13	5.28	4.92	5.11	0.15
40 Paris	5.57	5.48	5.42	5.49	0.06
41 Berlin	4.78	5.26	4.67	4.90	0.26
44 Svanvik	7.52	7.40	6.96	7.30	0.24
45 Chaumont	6.60	6.90	6.56	6.68	0.15
50 Katowice	8.95	9.18	9.11	9.08	0.09
51 Athens	4.47	4.34	4.84	4.55	0.21
52 Riga	5.09	4.77	4.91	4.92	0.13
53 Vienna	5.96	5.95	5.93	5.95	0.01
54 Sofia	missing	missing	missing	missing	missing
55 St. Petersburg	6.84	7.31	7.46	7.20	0.26

a) arithmetic mean value, b) SD = standard deviation

In figure 3 the corrosion loss values of zinc for the last and the recent trend exposure period are shown (For test sites 27, 35 and 54 values for the recent exposure period are missing, test site 55 is new whereas no values for the former trend exposure are given). It could be seen from the calculated differences between mass loss of period 2008/09 and

2011/12 that for 14 of 21 test sites a decrease of corrosion loss for the recent period could be found, 3 of 21 test sites show a very small increase and only 4 of 21 test sites have a larger increase.

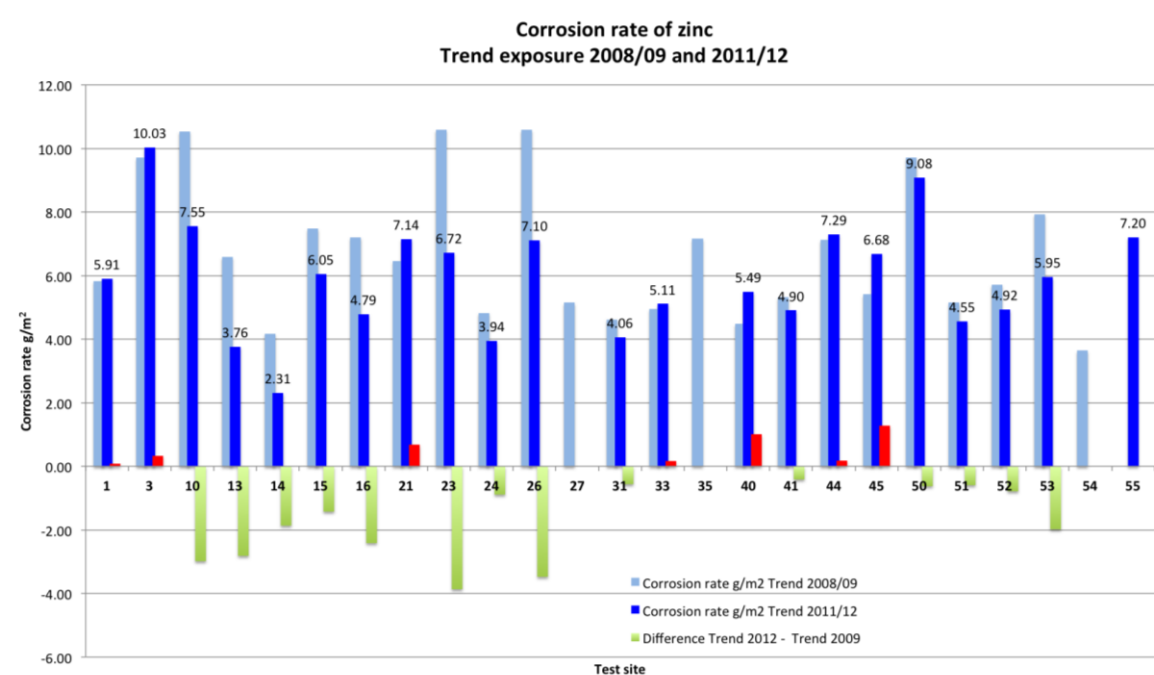


Figure 3 Comparison of loss of zinc (CH) trend exposure 2008/09 (first column) and 2011/12 (second column) and difference (negative values: decrease of corrosion rate of the recent exposure period)

In table 3 all corrosion loss data of zinc (CH) are listed. The box plot diagram for the last five exposure periods are shown in figure 4. It can be seen that there is a tendency of decrease for all corrosion loss values in the last three exposure periods although this decrease is relatively small especially by comparing the both last trend exposures.

Table 3 Comparison of corrosion loss of zinc (CH) in different exposure periods in open atmosphere

Test site	1 year	1 year	1 year	1 year	1 year	1 year	1 year	1 year	1 year	2 years	2 years	4 years	4 years
	1990-91 ^a	1992-93 ^a	1994-95 ^a	1997-98 ^b	2000-01 ^b	2002-03	2005-06 Trend	2008-09 Trend	2011-12 Trend	1987-89 ^a	1997-99 ^b	1987-91 ^a	1997-01 ^b
	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²
1	7.0	7.7	5.6	6.66	7.43	11.33		5.82	5.91	26.1	9.27	45.4	16.94
3	11.5	11.6	12.1	10.13	11.22	7.21	16.17	9.70	10.03	29.6	15.29	57.4	26.62
10	10.6	15.2	7.8	12.08	15.7	10.33	10.85	10.54	7.55	20	21.08	32.2	41.7
13	9.7		3.4	8.4	4.5	7.53	9.63	6.58	3.76	11.8	10.8	30.7	
14	9.9		3.1	7.65	3.8	8.63	5.50	4.18	2.31	14.3	10.75	29.5	22.63
15	12.1		5.5	10.36	10.32	7.01	7.14	7.48	6.05	23.1	10.7	42.8	27.51
16	7.6		6.1	6.69	5.43			7.20	4.79	18.3	9.68	34.9	18.68
21	5.6	6.6	3.5	4.73	6.2	9.49	12.55	6.46	7.14	16.2	9	25.4	19.13
23	8.4	10.5	5	7.77	21.33	15.76	19.54	10.58	6.72	20.5	14.04	27.4	29.27
24	6	4.5	4.2	7.6	5.95	9.74	8.04	5.15	3.94	23.2	12.38	26.7	24.52
26	6.7	4.8	6	7.73	9.03	14.59	16.35	10.58	7.10	9	12.76	18.6	20.13
27	12.3	10.6	7	11.67	8.46	11.02		5.15		17	19.98	27.9	35.4
31	4.8	3.5	2.3	4.51	4.83	11.83	3.76	4.62	4.06	9.6	7.95	15.7	17.75
33	3.9	4.7	1.7	4.56	2.67	6.70	6.67	4.94	5.11	4.5	7.61	10	18.58
35	9.4			7.27	7.66	10.54	7.36	7.15		9.5	10.44	17.1	21.74
40				11.34	5.94	8.10	6.44	4.48	5.49		19.67		39.03
41				6.48	7.22	10.39	12.10	5.32	4.90		13.26		25.12
44				6.32	9.27	6.12	6.79	7.12	7.29		12.65		22.36
45				4.23	9.04	7.08	14.44	5.41	6.68		9.05		17.32
50					16.22	11.80	12.03	9.71	9.08				
51						5.81	7.05	5.15	4.55				
52							10.13	5.71	4.92				
53								7.93	5.95				
54								3.64					
55									7.20				

^{a)} Data for the periods 1990-91, 1992-93, 1994-95 were extracted from ICP Materials report 29; ^{b)} 1987-91 from report 22; 1997-2001 from report 43.

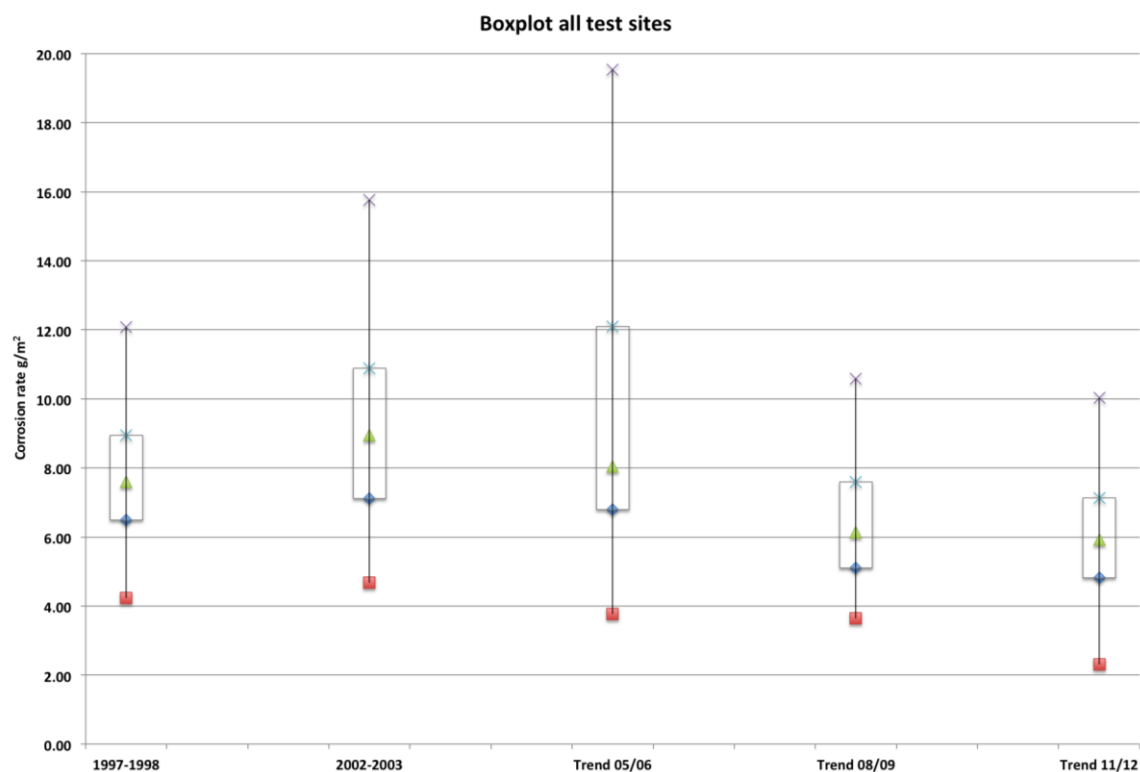


Figure 4 Boxplot of corrosion loss of zinc for the last five 1 year exposure periods

Analysis of corrosion products (extraction)

Aim

The analysis of anions in the corrosion products which were formed during exposure shows traces of the corrosive media which attacked the metal. The results are an indication which kind of substances was active in the corrosion. A quantitative statement is not possible because the extent of water-soluble salts that were washed away by rain events before the end of exposure is not known. Some of the anions can also be fixed to water-insoluble basic zinc hydroxides and will not be detected by the extraction method used. For this reason the results from different test sites and different exposure periods are not comparable and cannot be compared with one another. The results can be used qualitatively as a hint what corrosive substance came into contact with the metal surface.

Method

One metal sample after exposure was placed on the bottom of a beaker with a diameter a little bit larger than the diagonal of the sample. 50 mL of deionized water was added to the beaker and the water and sample was weakly agitated to let the water reach the entire sample surface. After 30 seconds of agitation the sample was turned and again agitated for further 30 seconds. The sample was rinsed on both sides and the washing solution was collected in the beaker. The sampled solution was transferred into a graduated flask. The

beaker was rinsed twice and the water was also collected in the flask. The flask was filled to 100 mL. This solution was analysed by capillary electrophoresis for inorganic anions and short carboxylic acid anions like formate and acetate. The values are referred to the surface of the specimen and given as microgram/cm².

Table 4 Results of analysis of water extracts of zinc corrosion products; Trend period 2011/12

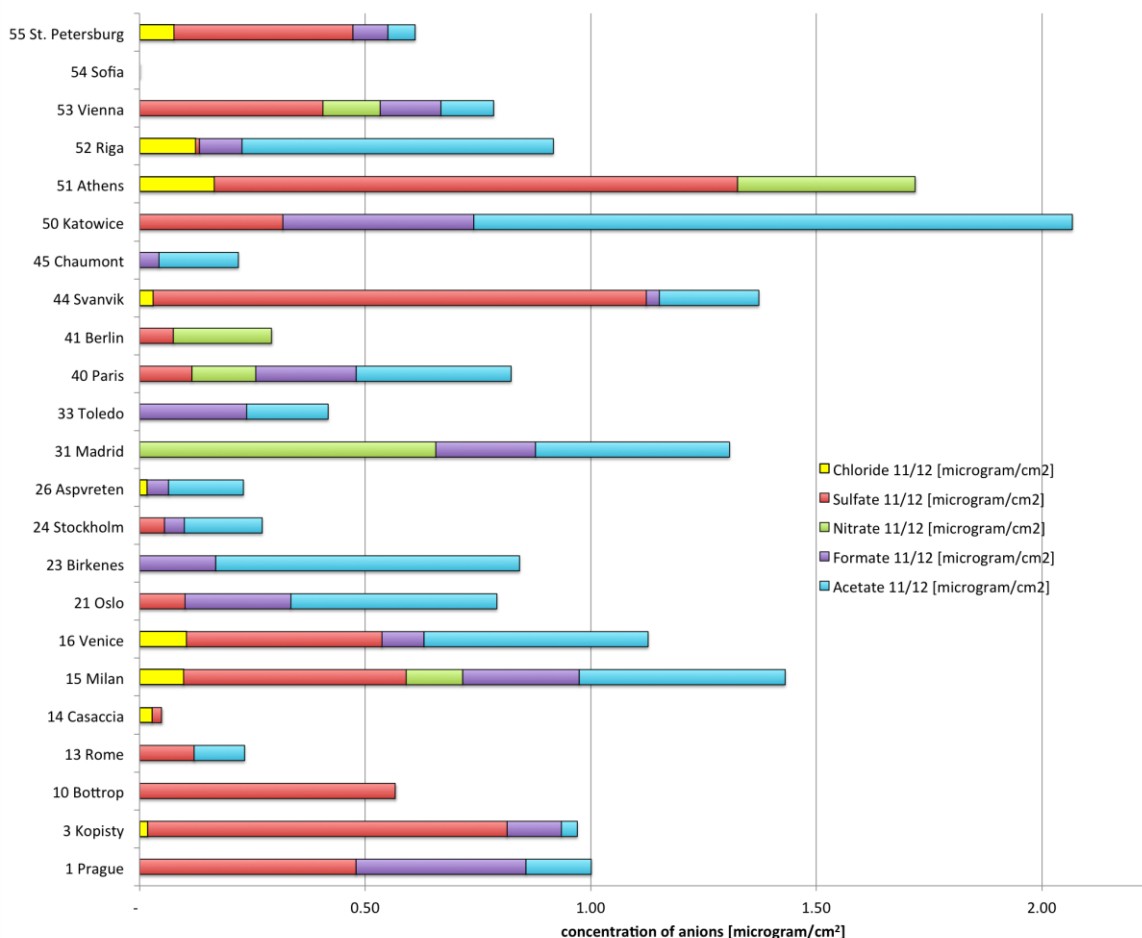
Test site		Ion concentration [$\mu\text{g}/\text{cm}^2$]				
		Cl ⁻	SO ₄ ²⁻	NO ₃ ⁻	Formate	Acetate
1	Prague	-	0.48	-	0.38	0.14
3	Kopisty	0.02	0.80	-	0.12	0.04
10	Bottrop	-	0.57	-	-	-
13	Rome	-	0.12	-	-	0.11
14	Cassacia	0.03	0.02	-	-	-
15	Milan	0.10	0.49	0.13	0.26	0.46
16	Venice	0.10	0.43	-	0.09	0.50
21	Oslo	-	0.10	-	0.23	0.46
23	Birkenes	-	-	-	0.17	0.67
24	Stockholm S.	-	0.06	-	0.04	0.17
26	Aspvreten	0.02	-	-	0.05	0.17
31	Madrid	-	-	0.66	0.22	0.43
33	Toledo	-	-	-	0.24	0.18
40	Paris	-	0.12	0.14	0.22	0.34
41	Berlin	-	0.08	0.22	-	-
44	Svanvik	0.03	1.09	-	0.03	0.22
45	Chaumont	-	-	-	0.04	0.18
50	Katowice	-	0.32	-	0.42	1.33
51	Athens	0.17	1.16	0.39	-	-
52	Riga	0.12	0.01	-	0.09	0.69
53	Vienna	-	0.41	0.13	0.13	0.12
54	Sofia	missing	missing	missing	missing	missing
55	St. Petersburg	0.08	0.40	-	0.08	0.06

*) - means $\leq 0.01 \mu\text{g}/\text{cm}^2$

Results

The analysis (table 4 and figure 5) shows that in some places chloride ions are involved in the corrosion process. In nearly all corrosion products water-soluble sulphate was detected. Only in some cases nitrates was found. Nearly all corrosion products contain formate and / or acetate. This can be seen as a hint that formic and acetic acid could have an influence on the corrosion processes of the exposed zinc samples.

Figure 5 Results of analysis of water extracts of zinc corrosion products; Trend period 2011/12



2.3 Copper

Copper samples were prepared and evaluated by the Main research Centre in Sweden (Swerea KIMAB). Plates of unalloyed copper (99%, DIN 1787) with dimensions 100 x 150 x 1 mm³ were exposed in triplicates per exposure site. The samples were degreased and weighed prior to exposure. The samples were received as per the instructions in Report 69: Technical manual for the trend exposure programme 2011-2012. Upon receiving the samples from the various locations the samples were checked for identity and sealed inside plastic bags with desiccant and held within a room of low humidity and controlled temperature.

The samples were then photographed in the “as-received” condition before any further analysis. Two samples from each site was analysed for mass loss and the third sample was kept for further analysis.

Weight change and mass loss after exposure are documented in this report. The samples were evaluated for mass loss by pickling away corrosion products according to ISO 8407.

There are no results for site no. 54 Sofia, Bulgaria as the samples were not returned to Swerea KIMAB for analysis.

Mass loss and weight change for individual sites are given in table 5 and presented in figures 6-7. The test sites which showed highest mass losses were Kopisty, Bottrop and Birkenes.

Table 5 Copper mass loss and weight change

Location	Mass loss 1987-88	Mass loss 1997-98	Mass loss 2011-12	Weight change 1987-88	Weight change 1997-98	Weight change 2011-2012
	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²	g/m ²
1 Prague, Czech Republic	10,83	5,52	5,23	0,68	0,14	0,38
3 Kopisty, Czech Republic	27,5	13,94	15,885	10,46	3,55	3,65
10 Bottrop, Germany	11,67	9,07	10,24	0,58	-0,05	1,26
13 Rome, Italy	6,7	5,97	4,675	1,06	0,63	0,735
14 Casaccia, Italy	13,43	6,74	4,835	2,16	0,72	0,71
15 Milan, Italy	7,81	5,63	3,51	-1,00	-0,13	0,98
16 Venice, Italy	16,41	7,47	4,05	3,66	0,89	0,61
21 Oslo, Norway	6,93	4,06	3,6	-0,85	-0,08	0,185
23 Birkenes, Norway	10,93	9,62	9,1	-0,27	0,42	0,635
24 Stockholm, Sweden	5,31	4,29	3,905	0,11	-0,06	0,095
26 Aspöreten, Sweden	10,67	5,77	3,555	1,11	0,39	0,13
31 Madrid, Spain	3,11	2,54	1,87	-0,41	0,25	0,145
33 Toledo, Spain	3,53	5,60	2,81	0,09	0,51	0,205
40 Paris, France	-	3,83	3,46	-	-0,02	0,36
41 Berlin, Germany	-	3,97	4,95	-	0,26	1,245
44 Svanvik, Norway	-	7,14	5,14	-	1,93	1,34
45 Chaumont, Switzerland	-	8,31	7,065	-	0,70	0,6
50 Katowice, Poland	-	-	5,315	-	-	-0,245
51 Athens, Greece	-	-	3,87	-	-	0,82
52 Riga, Latvia	-	-	4,68	-	-	0,56
53 Vienna, Austria	-	-	4,68	-	-	0,435
55 St Petersburg, Russia	-	-	4,28	-	-	0,5

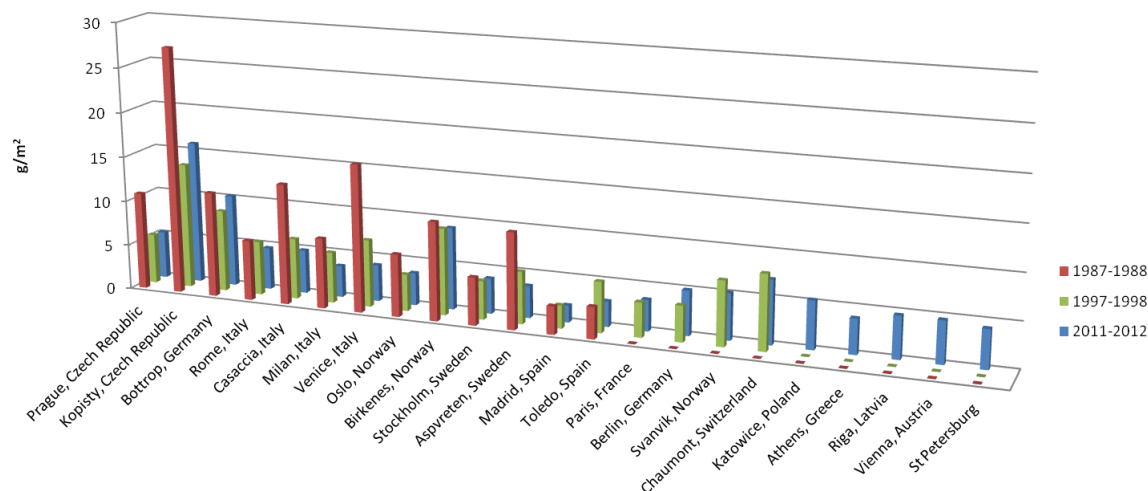


Figure 6 Mass loss per site per trend period.

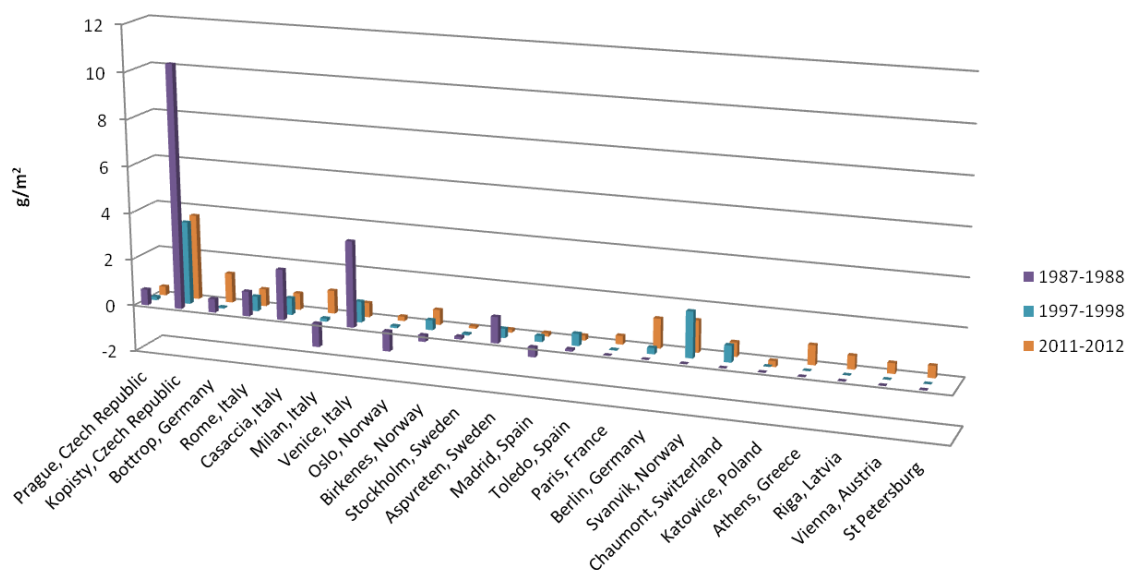


Figure 7 Weight change per site per trend period.

2.4 Weathering steel

Weathering steel samples were prepared and evaluated by the sub-centre in Spain (National Centre for Metallurgical Research (CENIM/CSIC)). The samples, with dimensions $100 \times 150 \times 1 \text{ mm}^3$, were obtained from commercial CORTEN-A type weathering steel sheets. Therefore, the chemical composition of the exposed samples (table 6) is similar to the composition of the weathering steel samples (ATMOFIX 52A) exposed in the period 1987-1995.

Table 6 Chemical composition of the weathering steel samples according to the producer.

C	Si	Mn	P	S	Cr	Cu	Ni
≤ 0.12	0.25-0.75	0.20-0.50	0.07-0.15	≤ 0.030	0.50-1.25	0.25-0.55	≤ 0.65

The samples, in triplicate per exposure site, were degreased and weighed prior to exposure. After the exposure, the samples were received as per the instructions in Report 69: Technical manual for the trend exposure programme 2011-2012. Upon receiving the samples from the various locations, the samples were checked for identity and sealed inside plastic bags with desiccant and held within a room of low humidity and controlled temperature.

The samples were then photographed in the “as-received” condition before any further analysis. Sample no. 3 from each exposure site was stored at low humidity and controlled temperature for surface and cross section characterization (SEM/EDS, X-Ray Diffraction, etc.), the results of which are not included in this report.

Mass loss after exposure is documented in this report. The samples were evaluated for mass loss by pickling away corrosion products according to ISO 8407. There are no results for site no. 54 Sofia, Bulgaria as the samples were not returned to CENIM for analysis.

<Continue here!!!>Mass losses for individual sites are given in table 7. They are also presented in figure 8, distributed from lowest to highest values. The test sites which showed highest mass losses were Kopisty, Bottrop and Katowice while the sites which showed the lowest were Toledo, Madrid and Aspvreten respectively. In figure 9, a comparison of mass losses values obtained in the periods (1987-1988) and (2011-2012) is shown.

Table 7 Weathering steel mass loss.

Location	Mass loss 1987-88 (g/m ²)	Mass loss 2011-12 (g/m ²)		
	ML _{average}	ML ₁	ML ₂	ML _{average}
1 Prague, Czech Republic	250	86.1	80.6	83.3
3 Kopisty, Czech Republic	304	192.2	199.0	195.6
10 Bottrop, Germany	257	193.7	194.4	194.1
13 Rome, Italy	154	82.7	85.6	84.2
14 Casaccia, Italy	185	109.5	110.1	109.8
15 Milan, Italy	279	86.5	88.0	87.2
16 Venice, Italy	193	93.3	87.7	90.5
21 Oslo, Norway	188	64.4	68.6	66.5
23 Birkenes, Norway	163	95.7	98.4	97.1
24 Stockholm, Sweden	238	65.2	67.1	66.1
26 Aspvreten, Sweden	132	46.2	49.0	47.6
31 Madrid, Spain	169	36.1	36.9	36.5
33 Toledo, Spain	55	24.3	24.3	24.3
40 Paris, France	-	70.7	73.0	71.9
41 Berlin, Germany	-	119.0	113.1	116.1
44 Svanvik, Norway	-	121.3	116.5	118.9
45 Chaumont, Switzerland	-	46.4	59.8	53.1
50 Katowice, Poland	-	165.4	169.6	167.5
51 Athens, Greece	-	83.3	82.2	82.8
52 Riga, Latvia	-	89.4	95.0	92.2
53 Vienna, Austria	-	76.2	76.6	76.4
55 St Petersburg, Russia	-	112.7	104.8	108.7

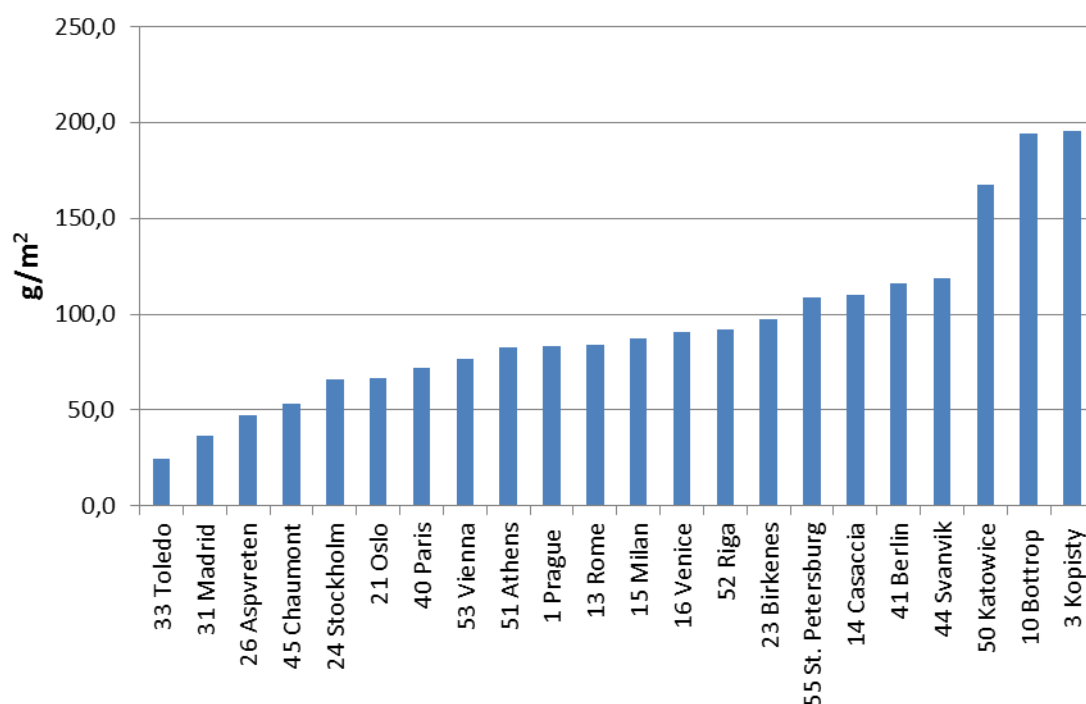


Figure 8 Mass loss (2011-2012), from lowest to highest.

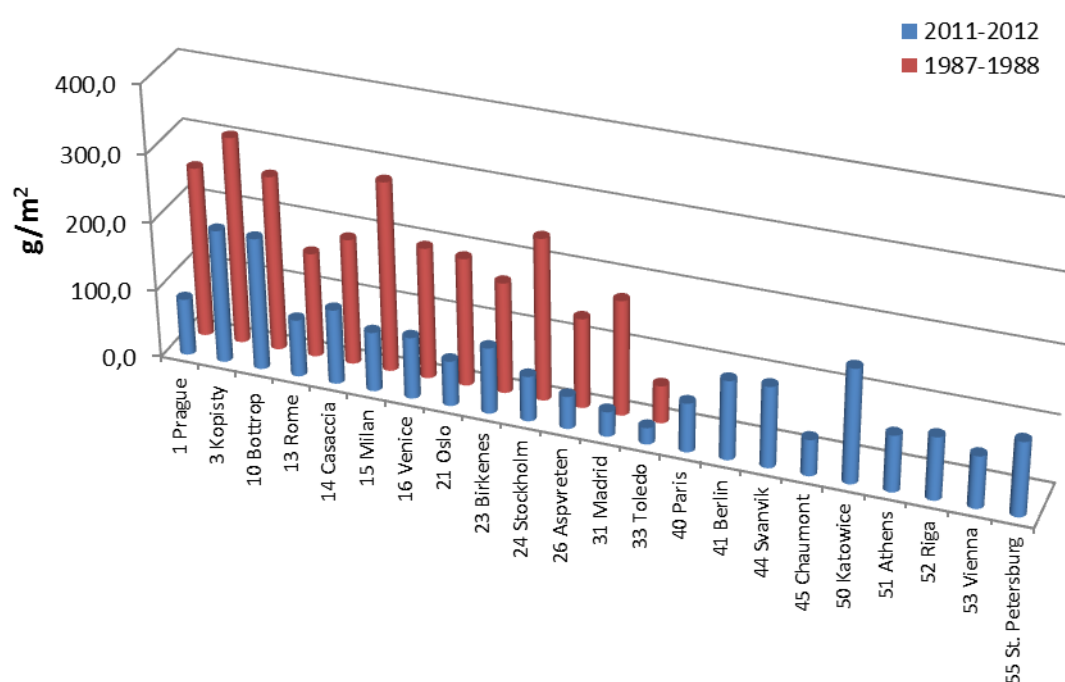


Figure 9 Mass loss, comparison of periods (1987-1988) and (2011-2012).

2.5 Limestone

Limestone samples were prepared and evaluated by the sub-centre in the UK (Building Research Establishment Ltd). Specimens of Portland limestone (Jurassic Oolitic limestone) with dimensions $50 \times 50 \times 8 \text{ mm}^3$ were used in triplicates per exposure site. The samples were washed, dried and weighed prior to exposure. The samples were received as per the instructions in Report 69: Technical manual for the trend exposure programme 2011-2018. Upon receiving the samples from the various locations the samples were checked for

identity and sealed inside plastic bags. The one year specimens were received back from 22 sites.

The samples were then removed from the bags, dried and weighed. They were then reweighed after 24 hours to confirm that they had reached constant mass.

When the 2005-06 were analysed the results seemed a little high at most sites and whether this was linked to the fact that a new batch of stone (from a different quarry though still from the same bed) had been used.

The 2008 stone was again a new batch – this time from the original bed and quarry. The results in 2008-09 were more consistent with the earlier trends – though it was noted that some of the less polluted sites were as high as in some of the earlier trend surveys which was thought to reflect the snow and cold in the winter of 2008-09. A comparison of the results for the different stone samples is shown in Table 8.

Surface recession at individual sites of limestone exposed in open atmosphere for 1 year exposure are summarised in Table 9.

Table 8 Comparison of surface recession (default), porosity and saturation coefficient data from the different samples of Portland Whit bed exposed over the period 1987- 2012.

Country	Site	1987 Stone	1997-2003	2005 Stone	2008 Stone	2008 Stone 2011-12 exposure	2013 Stone
Czech Republic	1. Prague	-4.73			-4.50	-7.24	
Norway	23. Birkenes	-14.78			-11.91	-13.33	
United Kingdom	27. Lincoln	-5.05	-4.55	-5.89	-6.78	ND	
	Porosity	22.40	16.20	18.85	23.60		23.70
	Saturation Coefficient	0.58	0.72	0.62	0.64		0.67
Switzerland	45. Chaumont	-7.06			-8.18	-11.98	
Poland	50. Krakow	-7.20			-11.25	-13.74	
Greece	51. Athens	-6.68			-8.65	-9.04	

Table 9 Surface recession data in μm for Portland limestone exposed in unsheltered conditions showing the comparison with the earlier trend data.

Country	Name	1987-88 μm Unsheltered	1997-98 μm Unsheltered	2002-03 μm Unsheltered	2005-06 μm Unsheltered	2008-09 μm Unsheltered	2011-12 μm Unsheltered	2005-12 Mean
01 Czech Rep	Prague-Bechovice	-22.90	-6.00	-5.33	-7.75	-4.50	-7.24	-6.50
03 Czech Rep	Kopisty	-19.40	-7.20	-7.91	-11.80	-11.11	-9.91	-10.94
05 Finland	Ähtäri	-15.00	-4.80	-5.46				
07 Germany	Waldhof Langenbrüge	-9.10	-3.30	-7.00				
08 Germany	Aschaffenburg	-8.50	-6.30	-7.62				
10 Germany	Bottrop	-17.40	-11.70	-11.88	-15.2	-8.56	-10.93	-11.56
13 Italy	Rome	-10.90	-6.00	-6.37	-8.87	-5.44	-7.75	-7.35
14 Italy	Casaccia	-6.70	-6.00	-6.71	-9.85	-5.93	-8.91	-8.23
15 Italy	Milan	-19.50	-9.00	-8.74	-15.09	-11.63	-9.54	-12.09
16 Italy	Venice	-8.80	-9.60	-7.17	-9.58	-6.98	-6.11	-7.56
21 Norway	Oslo	-11.70	-8.40	-7.31	-10.67	-4.32	-4.65	-6.55
23 Norway	Birkenes	-17.00	-7.50	-7.66	-14.65	-11.91	-13.33	-13.30
24 Sweden	Stockholm South	-14.00	-8.10	-7.29	-12.78	-9.57	-10.98	-11.11
26 Sweden	Aspvreten	-6.80	-3.60	-4.30	-7.79	-6.87	-8.25	-7.64
27 United Kingdom	Lincoln Cathedral	-9.90	-9.90	-7.63		-6.78		-6.78
31 Spain	Madrid	-13.30	-3.30	-3.91	-8.11	-2.64	-6.87	-5.87
33 Spain	Toledo	-6.20	-7.50	-5.82	-11.48	-7.49	-6.81	-8.59
34 Russia	Moscow	-9.40	-9.00	-7.94				
35 Estonia	Lahemaa	-9.90	-5.40	-8.91	-10.65	-8.33		-9.49
36 Portugal	Lisbon	-6.90	-11.40	-10.59				
37 Canada	Dorset	-8.90	-6.30		-14.31			-14.31
40 France	Paris	N/A	-8.70		-10.93	-8.14	-6.77	-8.61
41 Germany	Berlin	N/A	-3.30	-2.00	-8.59	-2.23	-3.00	-4.61
44 (73&76) Norway	Svanvik	N/A	-6.60	-9.11	-9.79	-12.79	-9.89	-10.82
45 Switzerland	Chaumont	N/A	-9.30	-9.04	-14.30	-7.06	-11.98	-11.11
46 United Kingdom	London	N/A	-7.50	-9.03				
51 Greece	Athens	N/A				-8.65	-9.04	
52 Latvia	Riga					-12.34	-15.04	
53 Austria	Vienna					-14.20	-10.24	
54 Bulgaria	Sophia					-5.35		
55 Russia	St.Petersberg						-8.36	

The results seem to be very consistent with those for 2008-09 – and in fact it does seem that the surface recession rates have been quite constant since 2005-06. The overall trend seems to be that rates of surface recession (and so of weathering) declined rapidly between 1987 and 1997 and since then the changes have been much smaller. This is illustrated in Figure 10.

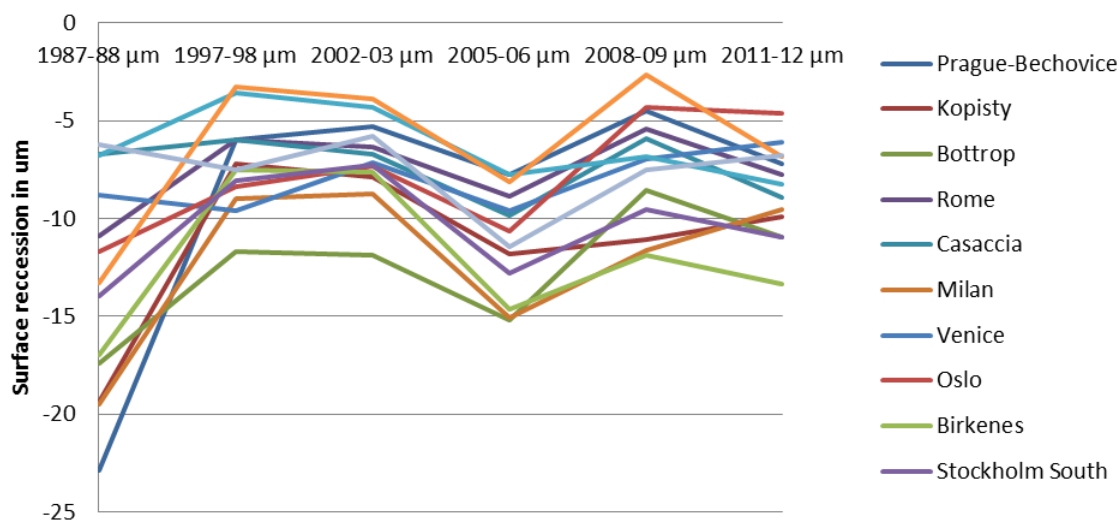


Figure 10 Trends in Portland limestone surface recession rates 1987-2012.

Two other effects seem to be apparent – the first is that the range has become less – the rate at most sites is similar ($9 \mu\text{m year}^{-1}$) and the second is that the change at the rural site has been less. Neither finding is too surprising as they reflect the reduction in urban sulphur dioxide.

3 Soiling

3.1 Modern glass

Modern glass was prepared and evaluated by the sub-centre in France (LISA).

Parameters for glass soiling for individual samples of modern glass exposed in sheltered atmosphere for 1 year exposure (2011/12) are summarised in Table 10. Figure 11 presents the mass of total deposited particles (TP/S in $\mu\text{g}/\text{cm}^2$) and Haze (in %), which is the most important parameter to assess the soiling.

During the 2011-12 trend exposure 22 samples were exposed, two of them (Katowice and Riga) were evidently exposed in horizontal position (instead of vertical) and thus their soiling is much higher than the other samples. It is also of note that the site for Paris has moved (from St-Eustache to Paris-Est Créteil University) which explain the difference of results compared to the previous campaigns.

Except these both samples, the highest levels of haze and TP/S were observed at the Paris, Athens and Casaccia test sites. The lowest soiling rate was found at the rural sites of Svanvik and Birkenes (for haze) and Aspvreten and Chaumont (in terms of deposited matter).

Data can be compared to the other campaigns (2005-06 and 2008-09) (Figure 12). The range of haze is similar for the different campaigns. The minimum value is 0.9% (Svanvik) for 2005-2006, 1.0% (Chaumont) for 2008-2009 and 1.0 % (Svanvik) for 2011-2012. The maximum value is found for Athens (8.9% for 2005-2006, 10.5% for 2008-2009 and 10.1% for 2011-2012 by excluding the new site of Paris). The average value (for sites with available data from the 3 campaigns) is very close for the 3 years of exposure: 2.9 ± 1.9 % (2005-2006) 3.0 ± 2.2 % (2008-2009) and 3.1 ± 2.3 % (2011-2012).

Considering the trends, haze significantly increases for Bottrop, Casaccia, Venice, Stockholm and Katowice and decreases for Prague, Paris and Vienna (Table 2). It stagnates for Kopisty, Rome, Milan, Oslo, Birkenes, Aspvreten, Madrid, Toledo, Lahemaa and Athens. For the other sites, the trend is not clear: the haze increases from 2005-2006 to 2008-2009 but decreases for the last year in Berlin and Svanvik and the haze decreases from 2005-2006 to 2008-2009 and increases for 2011-2012 in Chaumont. Finally, considering all the sites, haze doesn't show a very clear trend of evolution. A majority of sites display similar values during the three campaigns. For each campaign, between 3 and 4 sites present a significant increase and between 2 and 4 sites a decrease.

Table10 Type of sites, exposure dates, TP/S (in $\mu\text{g}/\text{cm}^2$) and haze results for the different sites

	Date deb	Date fin	jours	TP/S ($\mu\text{g}/\text{cm}^2$)	Haze (%)
Prague Lethany 1	18-10-11	13-10-12	361	4.6	1.4
Kopisty 3	19-10-11	22-10-12	369	8.8	2.5
Bottrop 10	04-10-11	07-10-12	369	7.0	2.0
Rome 13	22-11-11	04-12-12	378	4.7	3.1
Casaccia 14	25-11-11	05-12-12	376	59.0	7.8
Milan 15	29-11-11	11-12-12	378	18.3	2.0
Venice 16	01-12-11	13-12-12	378	18.7	4.9
Oslo 21	06-10-11	11-10-12	371	6.3	1.7
Birkenes 23	11-10-11	16-10-12	371	2.7	1.2
Stockholm 24	14-10-11	15-10-12	367	7.3	2.9
Aspvreten 26	17-10-11	18-10-12	367	1.7	1.6
Madrid 31	03-10-11	10-10-12	373	6.0	2.7
Toledo 33	05-10-11	01-10-12	362	8.8	2.4
Paris 40	07-10-11	01-10-12	360	60.2	10.1
Berlin 41	29-09-11	01-10-12	368	24.8	3.6
Svanvik 44	13-10-11	18-10-12	371	6.5	1.0
Chaumont 45	11-10-11	10-10-12	365	1.7	1.7
Katowice 50	20-10-11	29-10-12	375	558.7	38.0
Athens 51	03-10-11	03-10-12	366	33.5	8.7
Riga 52	26-10-11	26-10-12	366	1233.8	54.6
Vienna 53	07-11-11	08-11-12	367	4.3	1.6
St. Petersbourg 55	11-10-11	10-10-12	365	10.2	1.6

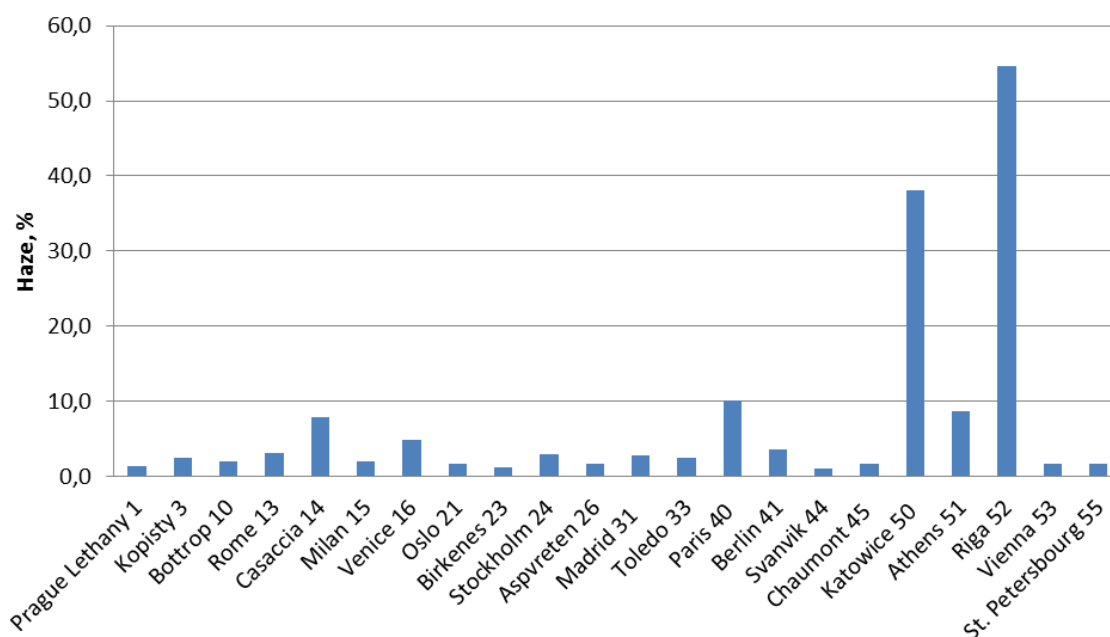


Figure 11 %Haze measured on modern glass samples exposed 1 year in sheltered condition.

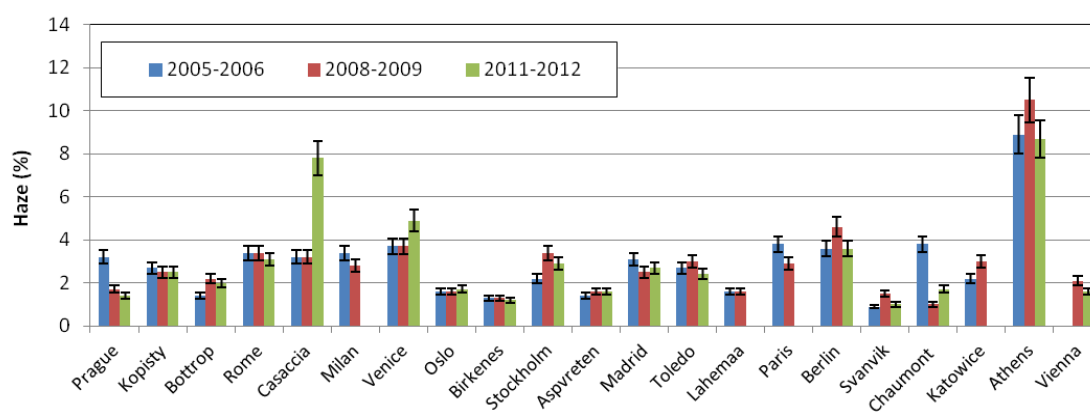


Figure 12 Comparison of haze for the different exposure campaigns.

4 Conclusions

- ICP Materials has completed the 2011-2012 trend exposure resulting in corrosion data on carbon steel, zinc and limestone and soiling data on modern glass for the network of test sites. Reporting of environmental data is proceeding according to plans and will be finalised in 2014.
- In addition to results for indicator materials, trend results for copper and weathering steel are available for the year 2011-2012 enabling further quantification of trends for additional materials.
- The soiling and corrosion results from the period 2011–2012 are similar to that in 2009-2010, in other words, no significant trend is observed. A more detailed analysis taking into account all trend results since 1987 including environmental data will be performed in 2014.

5 References

1. <http://www.unece.org/env/lrtap/WorkingGroups/wge/welcome.html>